

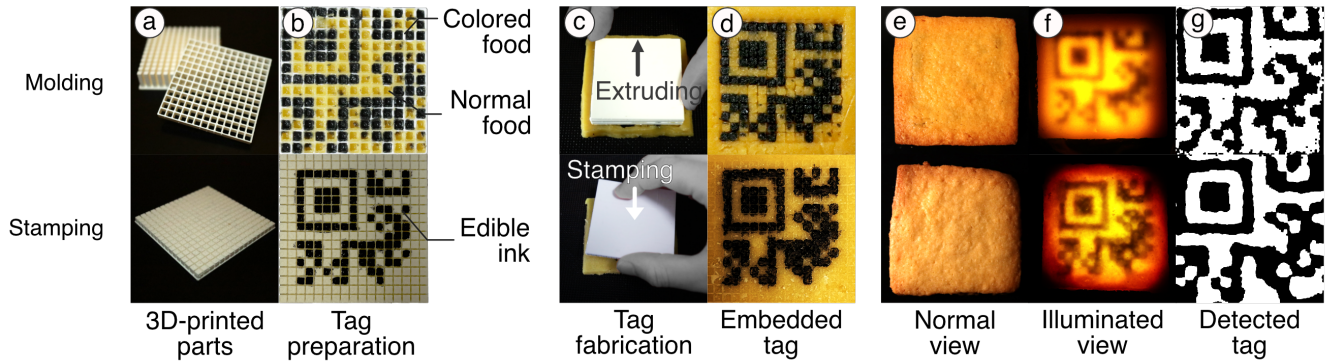
# An Exploratory Study on Fabricating of Unobtrusive Edible Tags

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**Figure 1: The overview of the proposed methods. (a – b) The 3D-printed mold and stamp prepared for tag embedding, (c – d) the embedding step, (e – f) tag visibility in normal and illuminated view, (g) extracted tags via image processing.**

## Abstract

Previous work in Human-Food Interaction has investigated how to embed tags into food. One promising approach involves controlling the internal structure using food 3D printers. However, this method is not widely accessible due to the current limitations of food 3D printers. This paper explores alternative fabrication techniques, molding and stamping, for embedding unobtrusive tags inside foods. Our preliminary evaluations showed that the proposed methods can embed tags and have the potential to employ a wide range of materials with lower costs compared with the food 3D printing technique. These findings suggest that low-cost edible tagging technologies could become more accessible and versatile.

## CCS Concepts

• Human-centered computing → Interaction techniques.

## Keywords

digital fabrication, human-food interaction, invisible tag

## ACM Reference Format:

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## 1 Introduction

Recent research in food tags has made them edible so that users can consume tags or reduce the non-edible packages for sustainability [Uji et al. 2017]. Specifically, previous work attempts to make the tag unobtrusive so that it does not reduce the dining experience as such when the tag is visible to the naked eye [Miyatake et al. 2022]. However, the previous method requires precise material properties such as the viscosity and size of particles due to using food 3D printing for fabricating tags.

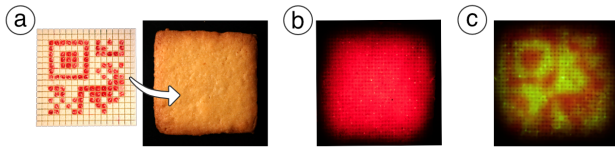
This work explores two new methods, molding and stamping, to identify a suitable approach for employing unobtrusive food tags across a wide range of materials compared to previous work that utilized a food 3D printer. The molding places multiple materials into desired tags, and the stamping draws a tag inside foods. We discuss the advantages, disadvantages, and initial evaluation that would allow the unobtrusive food tags to be embedded in a wider range of foods. Our research pushes the boundaries of conventional methods, aiming to transform food tag integration into a more sustainable and versatile solution.

## 2 Proposed Method

Figure 1 shows the overview of the proposed method and the results. We detail our proposed fabrication methods, molding and stamping in this section.

**Molding.** Molding, which involves filling material into a mold to create a desired shape, is commonly used in both the food industry and daily kitchens.

Our method uses molding techniques to create unobtrusive edible tags. We prepare a mold with two parts: a base with multiple cell-shaped holes and an extruder with protruding cells that fit into these holes to embed tags inside foods. The molds can be printed



**Figure 2: Use of other colors for tags. (a) Tag embedding using edible red ink. The tag is invisible in normal view. (b) The tag is still invisible under red illumination. (c) The tag becomes visible under green illumination.**

using a standard 3D printer. We fill the mold cells with normal food material (e.g., cookie dough) for the “0” parts and food material of different colors (e.g., blackened dough) for the “1” parts of a tag. The extruder positions the tag in the mold, and the tags are covered with a top layer to conceal the tag (Figure 1 (a)–(d) Top).

**Stamping.** Drawing on food surfaces with edible ink is common in daily cooking for intricate decorations on icing cookies or personalized messages on chocolates. Inspired by this technique, we enabled data embedding by stamping edible ink inside the food. We use a 3D-printed stamp with a guide for cells. To embed a tag, we paint the cells with black edible ink for the “1” parts of a tag. Then, we stamp the area to embed the tag, transferring the ink to the food’s surface and covering it with a top layer to conceal the tag from normal view (Figure 1 (a)–(d) Bottom).

### 3 Preliminary experiment

We conducted a preliminary experiment to verify the embeddability and detectability of our proposed method. We printed mold and stamp for data embedding using a standard 3D printer (Ulti-maker3) with 13 cells, each 3.6 mm in size, a total of 55.2 mm for each method. To make samples, we used cookie dough made of flour, sugar, egg yolk, and butter in a 1.0:0.5:0.2:0.5 ratio, which is empirically unsuitable for 3D food printers due to its hardness. We embedded the data by molding and stamping methods into the dough rolled out to an even thickness of 5 mm. The dough with the data was covered with a 1 mm layer of another cookie dough to conceal it from normal view and baked in an oven for 20 minutes.

We confirmed that the tags emerge when the cookie is illuminated from the backside but remain invisible to the naked eye without illumination (Figure 1 (e)–(f)). To check the detectability, we placed a camera on the top and a light on the bottom of the sample and experimented with simple image processing: denoising (Gaussian blur), contrast boosting (CLAHE), and binarization. We confirmed that these processed tags can be read by off-the-shelf tools such as QRQR (Denso Wave, Figure 1 (g)). The result found that the embedded tags with the proposed methods can be detected through commercially available tools.

We found that embedded tags become slightly visible to the naked eye even without light illumination when the cover layer is too thin. We additionally explored appropriate color combinations for the tag and illumination to reduce surface contrast by the tag in normal view while retaining detectability during illumination. We confirmed that the visibility of tags to the naked eye can be

**Table 1: Comparison of fabrication methods**

|                      | 3D-printing | Molding | Stamping |
|----------------------|-------------|---------|----------|
| Accessibility        | ▲           | ✓       | ✓        |
| Variety of materials | ▲           | ✓       | ✓        |
| Robustness of tags   | ✓           | ▲       | ▲        |

reduced while still detectable when using a red tag and green illuminations for cookies (Figure 2). Future work should explore other color combinations and the use of special materials such as quinine which fluoresces under ultraviolet light.

### 4 Discussion

Table 1 shows our results that assesses the feasibility of the proposed methods and their potential compared to the 3D printing method.

**Accessibility.** The proposed methods can be accessible to users as they only require 3D-printed parts. Users can easily integrate these methods into daily cooking routines since they utilize existing cooking techniques. Future work should involve measuring user experience through subjective experiments.

**Variety of food materials.** The proposed methods would allow data embedding in a variety of foods, unlike the food 3D printing method, which requires materials to be extruded from nozzles. Hence, they would extend the use case of data embedding in real-world applications. Future work should include exploring various materials with different properties.

**Robustness for tags.** We expect the error rate for data in the new methods to be higher than that of the 3D-printing method, as they involve manual steps for data embedding, which may introduce random errors. Future work should include quantitative experiments to measure error rates of manual procedures.

### 5 Conclusion

We present two data embedding methods, molding, and stamping, to achieve a low cost and wide material applicability. Our experiment confirmed the usability of the proposed method of embedding and acquiring the data using an off-the-shelf tag reader. We discussed our method compared to the prior fabrication method such as food 3D printing.

### Acknowledgments

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